

0 5 . 2

The student wanted to present the results for the distribution of plant heights in a histogram.

Figure 6 shows the outline of a results table.

Use **Figure 6** to design a suitable results table the student should use for a histogram of plant height measurements ranging between 60 cm and 120 cm.

[2 marks]

Figure 6

Question 5 continues on the next page

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0 5 . 3

A student investigated biodiversity in a community of bee species in a field of wild flowers.

The student trapped and identified the bee species in the field over 6 months.

Table 4 shows some of the student's results.

Table 4

Bee species	Total number of trapped bees after 6 months
Cuckoo bee	250
Domestic honeybee	800
Clover bee	200

The index of diversity in a community of organisms is determined using the formula

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

where N = total number of organisms of all species
and n = total number of organisms of each species.

Use the formula and information in **Table 4** to calculate the index of diversity in this community of bee species.

[2 marks]

Answer _____



0 5 . 4

In a different investigation, the student found that:

- the species richness of plants in a field of wild flowers is greater than the species richness of plants in a field of wheat
- the species richness of **bees** in a field of wild flowers is greater than the species richness of **bees** in a field of wheat.

Suggest **one** reason why the species richness of bees in a field of wild flowers is greater than the species richness of bees in a field of wheat.

Explain your answer.

[2 marks]

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